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Melatonin for prevention of metabolic side-effects of olanzapine in patients with first-episode schizophrenia: Randomized double-blind placebo-controlled study

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ABSTRACT

We aimed to determine the efficacy of melatonin 3 mg/day in prevention of olanzapine-induced metabolic side-effects. In a randomized double-blind placebo-controlled study, 48 patients with first-episode schizophrenia who were eligible for olanzapine treatment, were randomly assigned to olanzapine plus either melatonin 3 mg/day or matched placebo for eight weeks. Anthropometric and metabolic parameters as well as psychiatric symptoms using The Positive and Negative Syndrome Scale (PANSS) were assessed at baseline, week 4, and 8. Primary outcome measure was the change from baseline in weight at week 8. Data were analyzed using *t*-test, Mann–Whitney *U* test, and mixed-effects model. Thirty-six patients had at least one post-baseline measurement. At week eight, melatonin was associated with significantly less weight gain [mean difference (MD) = 3.2 kg, $P = 0.023$], increase in waist circumference [MD = 2.83 cm, $P = 0.041$] and triglyceride concentration [MD = 62 mg/dl, $P = 0.090$ (nearly significant)] than the placebo. Changes in cholesterol, insulin, and blood sugar concentrations did not differ significantly between the two groups. Patients in the melatonin group experienced significantly more reduction in their PANSS scores [MD = 12.9 points, $P = 0.014$] than the placebo group. No serious adverse events were reported. To summarize, in patients treated with olanzapine, short-term melatonin treatment attenuates weight gain, abdominal obesity, and hypertriglyceridemia. It might also provide additional benefit for treatment of psychosis. The study was registered in the ClinicalTrials.gov (Registration number: NCT01593774).

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1. Introduction

Despite well-established efficacy and lower incidence of extrapyramidal side effects, atypical antipsychotics (AAPs) are associated with high risk of weight gain, metabolic syndrome, diabetes mellitus, and hyperlipidemia (Coccurello and Moles, 2010; Pramyothin and Khaodhiar, 2010). This side-effect profile is particularly troublesome in patients with schizophrenia who are more likely to have other risk factors for developing cardiovascular diseases (Brown et al., 1999). Furthermore, excessive weight gain caused by AAPs can cause subjective distress, decreased self-esteem and noncompliance (Coccurello and Moles, 2010). In recent years, several management strategies have been proposed to reduce the metabolic burden of AAPs. Among many drugs that have been used to prevent AAP-associated metabolic disturbances are metformin, nizatidine, topiramate, and fluoxetine (Baptista et al., 2008). Contradictory results along with occasional intolerable side-effect profile of these drugs necessitate searching for more tolerable and efficacious options.

Melatonin, a multifunctional hormone, is secreted by pineal gland as well as several other cell types in the body (Korkmaz et al., 2009). Its wide range of activities encompasses scavenging free

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radicals, modulation of sleep, and regulation of energy metabolism and fat distribution (Raskind et al., 2007). By binding to its nuclear receptors, melatonin might also alter cell function through epigenetic processes (Korkmaz et al., 2009). Melatonin seems to be an important metabolic regulator through its widespread interactions with several related pathways in the body. In animal models of obesity and/or metabolic syndrome, melatonin prevented weight gain (Prunet-Marcassus et al., 2003; Raskind et al., 2007; Wolden-Hanson et al., 2000), attenuated obesity-related changes (Hussein et al., 2007; Wolden-Hanson et al., 2000), and improved the metabolic profile (Hussein et al., 2007; Wolden-Hanson et al., 2000). In high fat-fed rats, melatonin attenuated weight gain, and reduced concentration of glucose, insulin, adiponectin, leptin, triglycerides and cholesterol (Rios-Lugo et al., 2010). In an eight-week study on rats, melatonin reduced olanzapine-induced weight gain and visceral adiposity (Raskind et al., 2007). Results of a small randomized study have shown that ramelteon (a melatonin receptor agonist) could significantly attenuate hyperlipidemia associated with AAPs intake (Borba et al., 2011).

Although melatonin has been used as a sleep-enhancing agent in schizophrenia, its effects on metabolic side-effects of AAPs in human have not been investigated yet. The aim of the present randomized trial was to assess the efficacy of melatonin on olanzapine-induced metabolic side-effects in patients with first-episode schizophrenia.

2. Materials and methods

2.1. Trial design and setting

This was a single-center, eight-week, randomized, double-blind, placebo-controlled, and parallel-group study. The study was conducted in an academic psychiatric hospital (Shafa Hospital, affiliated with Guilan University of Medical Sciences, Rasht, Iran) from May 2012 to March 2013.

2.2. Participants

Male and female patients aged 18–65 years with diagnosis of schizophrenia based on Diagnostic and Statistical Manual (DSM)-IV-TR were recruited for the study. Diagnosis was based on Structured Clinical Interview for DSM-IV-TR Axis I Disorders (SCID-1) (First et al., 2002) and a senior psychiatrist (MJM) confirmed the diagnosis. All follow-up visits were done by a third-year psychiatric resident (PH) who was adequately trained to use the assessment instruments. The patients were required to be in their first-episode and to be eligible for starting olanzapine. Exclusion Criteria were: married women who were at reproductive age (unless they used a reliable non-hormonal contraception method), history of taking olanzapine in the recent three months, history of allergy or intolerance to olanzapine, history of significant head trauma (causing loss of consciousness more than 5 min or neurological or cognitive sequels), liver and kidney impairment, symptomatic cerebrovascular or cardiovascular disease, diabetes mellitus, metabolic syndrome (based on National Heart, Lung, and Blood Institute/American Heart Association definition), cancer, use of antiepileptic, antihypertensive, anticoagulant, or anti-platelet drugs, using inhibitors or stimulants of hepatic isoenzymes that metabolize melatonin or olanzapine (e.g. omeprazole. rifampin, fluvoxamine, ciprofloxacin, carbamazepine, modafinil), delirium, need for administration of other antipsychotics, and addictive disorders. The study protocol was approved by the Ethics Committee and the Institutional Review Board (IRB) of Guilan University of Medical Sciences (Approval Number: 190015211). After complete description of the study to the subjects, written informed consent was obtained from the participants and their legally authorized representatives. The trial was conducted in accordance with the last revision of the Declaration of Helsinki and was registered in the ClinicalTrials.gov (Registration number: NCT01593774).

2.3. Interventions

Olanzapine was started at 5 mg/day and was titrated up to 25 mg/day (increasing the dose by 5 mg every 3–7 days based on the response to treatment and patient's toleration). Participants were randomly assigned to melatonin 3 mg (Melatonin, Nature Made, CA, USA) tablet at 9:00 pm or placebo (Sobhan Pharmaceuticals, Rasht, Iran) with the same appearance as melatonin at 9:00 pm for eight weeks. The patients were hospitalized for the first few weeks and then were followed up in the outpatient clinic. In the hospital, taking medication was directly observed by the nurse. After discharge, compliance was assessed by counting the pills returned. All patients received clonazepam 2 mg at night for sleep enhancement. No other medication was allowed.

2.4. Primary and secondary outcomes

Anthropometric and biochemical parameters were assessed at baseline, and weeks 4 and 8. Anthropometric measures included weight, height, body mass index (BMI), waist circumference (WC), hip circumference (HC), and waist to hip ratio (WHR). WC was measured midway between the lower rib margin and iliac crest. HC was measured at the level of the greater trochanter. Blood pressure was also measured at similar intervals. Biochemical parameters included fasting blood sugar (FBS), fasting plasma insulin (baseline and week 8 only), fasting lipid profile (triglyceride (TG), high and low density lipoproteins (HDL, LDL), and total cholesterol). All lab tests were carried out by an experienced technician (who was aware of the study) in a single laboratory. Serum glucose concentrations were measured using oxidase method. Cholesterol, TG, LDL, and HDL levels were measured by enzymatic Kinetic Kit (DAX-48, Bayer Diagnostics, Germany). Fasting plasma insulin was measured using radioimmunoassay (Diagnostic System Lab. Inc Texas, USA). Homeostatic model assessment of insulin resistance (HOMA-IR) was calculated as [fasting insulin (μU/ml) × fasting glucose (mg/dl)/405]. Psychiatric symptoms were assessed using the Positive and Negative Syndrome Scale (PANSS) at baseline, week 4, and 8 (Kay et al., 1987). Primary outcome measure was the change from baseline in weight at week 8. Secondary outcome measures were changes in all the other variables.

2.5. Randomization, allocation concealment and blinding

Randomization list was prepared using a computer random number generator in a 1:1 ratio and block size of four. Treatment allocation was concealed from the study participants and the physicians who rated them using sequentially numbered and opaque envelopes. Treatment allocation and clinical evaluation of the patient were done by separate persons. The patients and their family members, the evaluator, the person responsible for administering intervention, and the statistician were blind to allocation.

2.6. Statistical analysis

Based on studies of other agents used for prevention of olanzapine-induced weight gain (particularly metformin) we assumed a final difference of three kilograms between the two groups to calculate sample size. Considering a two-sided significance of 5%, and a power of 80%, a sample size of 32 ($n = 16$ in each

group) was obtained. Considering a 20% drop-out rate, a final sample size of 40 was calculated.

SPSS version 15.0.0 (SPSS, Chicago, IL) was used to analyze the data except for mixed-effects model analysis (xtmixed command) and multiple imputation (mi command) for which STATA version 12.0 (StataCorp LP, College Station, TX) was used. All analyses were carried out on the data from the participants with at least one post-baseline measurement. To address the missing data we used both mixed-effect model analysis and multiple imputations. Xtmixed is the STATA's module for analyzing repeated measure data which can account for unbalanced data. For multiple imputations, we created ten multiple imputation datasets followed by using MI estimate command to calculate the average of the estimates across the multiple imputations. A P value of <0.05 was considered statistically significant.

3. Results

3.1. Participants

Forty eight patients were randomly assigned to receive melatonin ($n = 24$) or placebo ($n = 24$). Six patients in each group dropped out prior to first post-baseline visit, and one further patient in each group was lost to follow-up after fourth week. Hence, data of 36 (75%) patients ($n = 18$ in each group) were eligible for ITT analysis. Supplementary Fig. 1 summarizes the reasons for drop-outs. To the best of our knowledge, none of the drop-outs were because of adverse events. Table 1 shows the baseline characteristics of the patients. Other than the metabolic side-effects, both groups tolerated medications well. Only one patient in the placebo group experienced dizziness at week 4. Compliance rates (>90% in both arms) did not significantly differ between the two groups. None of the patients experienced serious side-effects or extrapyramidal symptoms.

3.2. Primary outcome

Baseline weight did not differ significantly between the two groups ($P = 0.822$). By week 8, patients in the melatonin group gained significantly less weight (3.3% of the baseline) than the placebo group (8.5% of the baseline) [MD (95%CI) = 3.2 (0.5–6.0) kg, $t(34) = 2.39$, $P = 0.023$]. A mixed-effects model for prediction of weight showed a significant time × treatment interaction (TTI) for both week 4 ($P = 0.033$) and week 8 ($P = 0.003$) (Table 3).

3.3. Secondary outcomes

3.3.1. Anthropometric measures and blood pressure

By week 8, patients in the melatonin group experienced less increase in BMI [MD (95%CI) = 1.07 (0.15–1.99) kg/m^2, $t(34) = 2.38$, $P = 0.024$] and WC [MD (95%CI) = 2.83 (0.12–5.54) cm, $t(34) = 2.13$, $P = 0.041$] than the placebo group. P values for joint TTI were 0.016 and 0.024 respectively.

Another outcome which was not defined *a priori* was the number of patients with clinically meaningful weight gain (defined as >7% of the baseline weight). Nine patients in the placebo group (50%) compared with four patients in the melatonin group (22%) experienced clinically significant weight gain by the end of the study ($\chi^2 = 3.01$, $P = 0.083$).

No significant difference was observed in changes of blood pressure, WHR, and HC between the two groups (Tables 2 and 3).

3.3.2. Biochemical parameters

Changes in values of insulin, FBS, HOMA-IR, HDL, and LDL were not significantly different between the two groups (Tables 2 and 3).

Table 1
Baseline characteristics of the patients.

Variable	Melatonin ($n = 18$)	Placebo ($n = 18$)
Gender, male (%)	13 (72%)	12 (67%)
Age, years, mean (SD)	32.7 (7.3)	32.8 (8.2)
Smokers (%)	11 (61%)	9 (50%)
Olanzapine final dose, n	15 mg/day ($n = 6$), 20 mg/day ($n = 6$), 25 mg/day ($n = 6$)	15 mg/day ($n = 1$), 20 mg/day ($n = 13$), 25 mg/day ($n = 4$)
Olanzapine final dose, mean (SD)	20.0 mg/day	20.8 mg/day
Duration of hospitalization, days, mean (SD)	27.6 (6.2)	28.4 (6.0)
PANSS, mean (SD)		
• Total score	113.5 (12.7)	103.5 (18.0)
• Negative symptoms	26.9 (5.4)	24.0 (5.5)
• Positive symptoms	28.4 (5.0)	24.7 (7.4)
• General psychopathology	58.2 (6.9)	54.8 (7.2)
Weight, kg, mean (SD)	65.7 (10.9)	64.9 (9.5)
Height, cm, mean (SD)	165.7 (7.9)	167.2 (8.1)
Body mass index, kg/m^2, mean (SD)	23.9 (3.7)	23.2 (3.2)
Waist circumference, cm, mean (SD)	78.1 (8.3)	75.2 (9.4)
Hip circumference, cm, mean (SD)	86.7 (7.9)	85.9 (7.3)
Waist/hip ratio, mean (SD)	0.90 (0.05)	0.87 (0.07)
Blood pressure, diastolic/systolic, mmHg, mean (SD)	114.7 (7.6)/ 72.5 (4.6)	115.3 (9.6)/73.0 (6.5)
Total cholesterol, mg/dl, mean (SD)	197.2 (39.4)	173.5 (45.0)
Low density lipoprotein, mg/dl, mean (SD)	126.6 (30.0)	107.5 (40.5)
High density lipoprotein, mg/dl, mean (SD)	39.9 (10.2)	40.3 (7.7)
Fasting triglyceride, mg/dl, mean (SD)	167.1 (76.0)	153.9 (57.6)
Fasting blood glucose, mg/dl, mean (SD)	85.4 (12.5)	81.7 (10.8)
Fasting plasma insulin, μU/mL, mean (SD)	5.5 (2.1)	8.9 (5.2)
HOMA-IR, mean (SD)	1.13 (0.44)	1.81 (1.09)

PANSS: The Positive and Negative Syndrome Scale, SD: standard deviation, HOMA-IR: homeostatic model for assessment of insulin resistance.

TG concentrations showed a near-significant difference in favor of the melatonin group [MD (95%CI) = 62 (−10 to 135) mg/dl, $t(34) = 1.75$, $P = 0.090$] at the end of the trial (P value of TTI = 0.045). At the end of the study, changes in total cholesterol levels did not significantly differ between the two groups, although the joint TTI interaction was significant ($P = 0.012$) mainly due to cholesterol reduction in the melatonin group at week 4.

3.3.3. PANSS

At the end of the trial, patients in the melatonin group achieved significantly more reduction in their PANSS total score [MD (95% CI) = 12.9 (2.8–23.0), $t(34) = 2.61$, $P = 0.014$] than the placebo group (P value of TTI = 0.006) (Tables 2 and 3). Only changes in general psychopathology subscale was significantly different between the two groups (P value of TTI = 0.002) (Tables 2 and 3).

4. Discussion

Our findings show that short-term melatonin treatment attenuates weight gain, abdominal obesity, and possibly hypertriglyceridemia induced by olanzapine in patients with first-episode schizophrenia. To the best of our knowledge, this is the first randomized study addressing role of melatonin in counteracting the metabolic adverse effects of olanzapine in human. Evaluating first-episode schizophrenic patients treated with olanzapine, not only made the study more homogeneous, but also let us

Table 2
Comparison of outcomes between the two groups.

Variable	Week 4, Mean (SEM)	Week 8, Mean (SEM)	Change from baseline at week 8, Mean (SEM)	*t* value	*P* value
Weight, kg					
• Melatonin	67.1 (2.54)	67.9 (2.70)	2.2 (0.60)	2.39	**0.023**[b]
• Placebo	68.5 (2.48)	70.4 (2.57)	5.4 (1.22)		
Body mass index, kg/m^2					
• Melatonin	24.4 (0.83)	24.7 (0.88)	0.8 (0.21)	2.38	**0.024**
• Placebo	24.5 (0.75)	25.1 (0.74)	1.9 (0.40)		
Blood pressure, mmHg					
• Melatonin	114.2 (1.95)/73.3 (0.90)	116 (1.40)/73 (0.80)	1.3 (1.2)/0.9 (0.8)	SBP: 0.54	SBP: 0.592
• Placebo	113.0 (1.57)/71.4 (1.55)	115 (1.45)/74 (0.90)	−0.2 (2.5)/3.5 (1.6)	DBP: 1.45	DBP: 0.157
Waist circumference, cm					
• Melatonin	78.8 (1.82)	79.2 (1.73)	1.1 (0.78)	2.13	**0.041**
• Placebo	78.0 (2.13)	79.2 (2.10)	3.9 (1.09)		
Hip circumference, cm					
• Melatonin	87.08 (1.69)	87.6 (1.72)	0.9 (0.78)	1.53	0.137
• Placebo	86.4 (2.58)	89.1 (1.84)	3.1 (1.31)		
Waist/hip ratio					
• Melatonin	0.90 (0.01)	0.90 (0.01)	0.004 (0.01)	0.66	0.513
• Placebo	0.91 (0.02)	0.89 (0.01)	0.014 (0.01)		
Total cholesterol, mg/dl					
• Melatonin	195.4 (7.34)	198.5 (10.76)	−1.5 (11.3)	1.56	0.130
• Placebo	212.5 (8.92)	202.5 (7.80)	23.9 (11.96)		
Low density lipoprotein, mg/dl					
• Melatonin	123.5 (5.08)	123.9 (7.11)	−2.7 (7.80)	0.85	0.402
• Placebo	127.4 (8.70)	116.3 (5.23)	8.8 (11.09)		
High density lipoprotein, mg/dl					
• Melatonin	44.9 (3.26)	40.9 (2.43)	1.0 (2.1)	0.37	0.716
• Placebo	42.6 (2.70)	42.7 (2.40)	2.4 (3.19)		
Triglyceride, mg/dl					
• Melatonin	192.9 (16.6)	168.2 (19.53)	1.1 (22.23)	1.75	0.090
• Placebo	246.1 (19.48)	217.5 (26.87)	63.6 (29.00)		
Fasting blood sugar, mg/dl					
• Melatonin	86.3 (1.23)	90.1 (1.48)	4.7 (3.51)	1.44	0.160
• Placebo	85.0 (2.36)	92.7 (1.77)	11.0 (2.58)		
Fasting insulin, μU/ml					
• Melatonin	–	11.4 (3.03)	5.9 (3.09)	$z = -1.292$[a]	0.205
• Placebo	–	10.2 (1.79)	0.9 (2.35)		
HOMA-IR					
• Melatonin	–	2.6 (0.77)	1.49 (0.77)	$z = -0.809$[a]	0.433
• Placebo	–	2.4 (0.46)	0.50 (0.54)		
PANSS, total					
• Melatonin	94.2 (2.93)	82.0 (4.35)	−31.6 (4.09)	2.61	**0.014**
• Placebo	87.7 (4.41)	84.9 (4.58)	−18.7 (2.78)		
PANSS, positive					
• Melatonin	21.4 (1.11)	18.0 (1.26)	−10.4 (1.09)	1.58	0.124
• Placebo	18.0 (1.13)	16.8 (1.10)	−7.9 (1.15)		
PANSS, negative					
• Melatonin	23.1 (0.83)	20.7 (1.13)	−6.2 (1.42)	1.67	0.104
• Placebo	21.2 (1.22)	20.7 (1.30)	−3.3 (1.00)		
PANSS, general					
• Melatonin	49.6 (1.51)	43.3 (2.32)	−14.9 (2.17)	2.84	**0.008**
• Placebo	48.5 (2.31)	47.3 (2.42)	−7.4 (1.48)		

PANSS: The Positive and Negative Syndrome Scale, SEM: standard error of mean, HOMA-IR: homeostatic model for assessment of insulin resistance.
[a] Due to severe skewness and impossibility of achieving a normal distribution (despite transformation) a Mann–Whitney test was used.
[b] Bold values represent *P* values < 0.05.

Table 3
Results of mixed-effects analyses.

Variable	Coefficient	Standard error	Z	$P > \|z\|$	95% confidence interval	P value for joint group × time[b]
Weight, kg						
• Melatonin group × week 4[a]	−2.2	1.0	−2.13	**0.033**[c]	−4.3 to −0.2	**0.009**
• Melatonin group × week 8	−3.2	1.1	−2.98	**0.003**	−5.3 to −1.1	
Body mass index, kg/m^2						
• Melatonin group × week 4	−0.7	0.3	−2.09	**0.036**	−1.4 to −0.04	**0.016**
• Melatonin group × week 8	−1.0	0.4	−2.92	**0.004**	−1.8 to −0.3	
Waist circumference, cm						
• Melatonin group × week 4	−2.0	1.1	−1.88	0.060	−4.1 to 0.1	**0.024**
• Melatonin group × week 8	−2.9	1.1	−2.65	**0.008**	−5.1 to −0.8	
Hip circumference, cm						
• Melatonin group × week 4	−0.1	1.7	−0.05	0.961	−3.5 to 3.3	0.275
• Melatonin group × week 8	−2.5	1.8	−1.42	0.155	−5.9 to 0.9	
Waist/hip ratio						
• Melatonin group × week 4	−0.03	0.02	−1.34	0.182	−0.08 to 0.01	0.385
• Melatonin group × week 8	−0.01	0.02	−0.35	0.726	−0.05 to 0.04	
Systolic blood pressure, mmHg						
• Melatonin group × week 4	1.7	2.8	0.59	0.557	−3.9 to 7.2	0.818
• Melatonin group × week 8	1.4	2.9	0.49	0.621	−4.2 to 7.1	
Diastolic blood pressure, mmHg						
• Melatonin group × week 4	−0.3	1.7	−0.16	0.870	−3.6 to 3.0	0.257
• Melatonin group × week 8	−2.6	1.7	−1.5	0.131	−6.0 to 0.8	
Total cholesterol, mg/dl						
• Melatonin group × week 4	−40.8	13.8	−2.96	**0.003**	−67.8 to −13.8	**0.012**
• Melatonin group × week 8	−22.3	14.0	−1.59	0.111	−49.8 to 5.2	
Low density lipoprotein, mg/dl						
• Melatonin group × week 4	−22.9	12.6	−1.82	0.069	−47.6 to 1.8	0.191
• Melatonin group × week 8	−11.6	12.8	−0.91	0.365	−36.8 to 13.5	
High density lipoprotein, mg/dl						
• Melatonin group × week 4	2.7	4.1	0.66	0.508	−5.3 to 10.8	0.575
• Melatonin group × week 8	−1.6	4.2	−0.39	0.698	−9.8 to 6.6	
Triglyceride, mg/dl						
• Melatonin group × week 4	−66.3	31.2	−2.12	**0.034**	−127.5 to −5.1	**0.045**
• Melatonin group × week 8	−69.3	31.8	−2.18	**0.029**	−131.6 to −7.0	
Fasting blood sugar, mg/dl						
• Melatonin group × week 4	−2.5	3.8	−0.66	0.510	−9.9 to 4.9	0.326
• Melatonin group × week 8	−5.7	3.8	−1.49	0.135	−13.1 to 1.8	
PANSS, total						
• Melatonin group × week 4	−3.5	3.6	−0.96	0.338	−10.6 to 3.6	**0.006**
• Melatonin group × week 8	−11.7	3.7	−3.14	**0.002**	−19.0 to −4.4	
PANSS, positive subscale						
• Melatonin group × week 4	−0.3	1.2	−0.22	0.822	−2.7 to 2.1	0.160
• Melatonin group × week 8	−2.2	1.3	−1.77	0.077	−4.7 to 0.2	
PANSS, negative subscale						
• Melatonin group × week 4	−0.9	1.3	−0.69	0.493	−3.4 to 1.6	0.158
• Melatonin group × week 8	−2.5	1.3	−1.90	0.057	−5.1 to 0.1	
PANSS, general subscale						
• Melatonin group × week 4	−2.3	1.9	−1.17	0.242	−6.2 to 1.6	**0.002**
• Melatonin group × week 8	−6.9	2.0	−3.42	**0.001**	−10.9 to −3.0	

PANSS: The Positive and Negative Syndrome Scale, HOMA-IR: homeostatic model for assessment of insulin resistance.

[a] In the first column "melatonin group × week 4" means the interaction of being in the melatonin group versus the placebo group × measurement at week 4 versus at baseline visit.

[b] The P value of the last column combines the P values from both interactions.

[c] Bold values represent P values < 0.05.

study one of the subgroups who are most susceptible to weight gain (Strassnig et al., 2007).

A review found that adding metformin (duration = 12 weeks, MD = 4.9 kg), reboxetine (duration = 6 weeks, MD = 1.6 and 3.0 kg), and topiramate (duration = 12 weeks, MD = 1.4 kg) to olanzapine could significantly prevent weight gain (Baptista et al., 2008). According to a newer study, zonisamide might also be effective in the prevention of olanzapine-induced weight gain (duration = 16 weeks, MD = 4.1 kg) (McElroy et al., 2012).

Consistent with our findings, melatonin has been shown to significantly decrease visceral fat pad and abdominal obesity in several animal models including olanzapine-treated rats (Hussein et al., 2007; Prunet-Marcassus et al., 2003; Raskind et al., 2007; Wolden-Hanson et al., 2000). In contrast, administration of ramelteon in patients taking AAPs showed no significant effect on weight in one study (Borba et al., 2011). The mentioned study was different from ours in that it investigated a heterogeneous group of AAPs in older patients and only those with the evidence of metabolic syndrome or obesity. Besides, neither group experienced significant weight gain during the study period. On the contrary, amount of weight gain in the placebo group of our study (five kg) is comparable to the findings of previous studies (Patel et al., 2009; Wu et al., 2008).

Melatonin reduced hypertriglyceridemia induced by olanzapine causing TG concentration to return to its baseline values. Importantly, among hyperlipidemias, taking olanzapine is associated with higher risk of hypertriglyceridemia than hypercholesterolemia (Atmaca et al., 2003). Moreover, in comparison with other AAPs, olanzapine is more likely to cause hypertriglyceridemia (Pramyothin and Khaodhiar, 2010). Ramelteon was shown to exert similar effects on AAP-induced metabolic disturbances, although it did not reach statistical significance (Borba et al., 2011). The evidence for the efficacy of melatonin in treatment of hypertriglyceridemia has been reported in some (Agil et al., 2011; Rios-Lugo et al., 2010) but not all studies (Kozirog et al., 2011; Rindone and Achacoso, 1997). In vitro melatonin administration may inhibit differentiation and hypertrophy of adipocytes (Alonso-Vale et al., 2009). This suggests that the weight-reducing effect of melatonin occurs through decreasing fat (and not lean body mass) content. With the regimen used in our study melatonin did not alter LDL and HDL levels significantly. However, olanzapine itself did not significantly alter the level of HDL and LDL either. Previous studies addressing the effect of melatonin on serum cholesterol profile have shown contradictory results (Agil et al., 2011; Hussein et al., 2007; Kozirog et al., 2011; Rindone and Achacoso, 1997). The causes of such discrepancies are unclear, although they might be partially explained by the differences in the studied models, underlying conditions, and the effect of dose and duration of treatment.

We observed no significant difference in insulin concentration and HOMA-IR between the two groups. In some animal models, melatonin suppressed insulin secretion through modulation of intracellular cyclic adenosine monophosphate (cAMP) levels (Peschke and Muhlbauer, 2010). In a study on old obese rats, melatonin treatment improved insulin sensitivity through increased insulin-induced phosphorylation of proteins, thereby enhancing glucose uptake. This effect occurred independent of weight loss (Zanuto et al., 2013). Studies in diverse groups of humans, however, have shown inconsistent findings (Cagnacci et al., 2001; Seabra et al., 2000). In a recent study of melatonin-modulated insulin release from human islet cells and mouse Min6 b-cells (Ramracheya et al., 2008), physiological concentrations of melatonin could induce insulin release at stimulatory and sub-stimulatory glucose concentrations in human cells. The authors suggested that this stimulatory effect was probably occurred indirectly through an increase in glucagon secretion. Interestingly, this was not the case in mouse, in which they found inhibitory effect of melatonin on insulin. Another study has demonstrated a positive correlation between melatonin and insulin in patients with metabolic syndrome (Robeva et al., 2008). The authors proposed that melatonin might increase insulin concentrations to compensate for insulin resistance. Taken together, melatonin effect on glucose metabolism in patients who receive AAPs can be an area of further investigation.

Several complex pathways might explain metabolic derangements caused by AAPs. These include increased appetite, decreased thermogenesis, interaction with several neurotransmitter receptors, modulation of neuropeptides, and disruption of circadian rhythm (Coccurello and Moles, 2010). Hypothetically, melatonin counteracts metabolic side-effects of AAPs by interfering in several of these pathways. Nonetheless, disruption of circadian rhythm might be the most intriguing to discuss. AAPs can cause variety of sleep disorders such as rhythm disturbances, restless leg syndrome, and night-eating syndromes (Cohrs, 2008). Interestingly, sleep deprivation causes hyperphagia in rats (Everson, 1995) and a relationship has been observed between obesity and sleep curtailment (Lucassen et al., 2012). In addition, circadian misalignment can affect hormones involved in the regulation of metabolic processes, e.g. insulin, leptin, and ghrelin (Ruger and Scheer, 2009). Therefore, one key explanation for melatonin action may be its counteraction of metabolic disturbances through regulation of circadian rhythm (Raskind et al., 2007). Significant circadian expression of genes in adipose tissue, relation of shift working to metabolic disturbances, and association between CLOCK gene mutation and obesity are a few of many important clues of the link between melatonin and metabolic perturbations (Reiter et al., 2012). In healthy adults, inadequate sleep delays circadian melatonin phase and increases energy intake in excess of energy expenditure (Markwald et al., 2013). Melatonin enhances oxidative phosphorylation in rat brain and liver mitochondria, thereby enhancing energy expenditure (Martin et al., 2002). Melatonin also interacts with the hormone leptin. Leptin concentration alters with circadian rhythm and is inversely related to daily fluctuations of melatonin concentrations (Coccurello and Moles, 2010). Expression of suppressors of cytokine signaling (SOCSs) can induce leptin resistance, thereby causing a hyperphagic effect (Tups et al., 2004). Interestingly, following brain contusion in Wistar rats, melatonin prevented overexpression of SOCS3, thereby attenuating brain damage (Tsai et al., 2011).

An unexpected finding in our study was the beneficial adjunctive effect of melatonin on psychotic symptoms. Importantly, differences in baseline characteristics, olanzapine dose, or duration of hospitalization do not explain this difference. To the best of our knowledge, no published study has directly addressed this effect of melatonin in human. However, disturbances in melatonin secretion pattern have been reported in schizophrenia (Anderson and Maes, 2012). A recent study have shown that melatonin (MT)1 receptor polymorphism is associated with schizophrenia (Park et al., 2011). Pineal calcifications have been linked to thought disorders in schizophrenia (Sandyk, 1993). Prepulse inhibition (PPI) of the acoustic startle reflex that is disrupted in schizophrenia can also be impaired by knocking out the MT1 receptor and pinealectomy (Uzbay et al., 2013; Weil et al., 2006). Furthermore, chronic melatonin treatment can improve PPI disruption (Uzbay et al., 2013). Finally, modulation of dopaminergic and serotoninergic pathways as well as anti-inflammatory and antioxidant actions of melatonin may contribute to its beneficial effect in treatment of schizophrenia (Anderson and Maes, 2012; Uzbay et al., 2013).

There are few studies regarding the use of melatonin in subjects with similar indications to our patients. Studies have used

melatonin in a range of 2.5–10 mg/day for treatment of patients with metabolic syndrome, hypertension, and nonalcoholic steatohepatitis (Gonciarz et al., 2013; Kozirog et al., 2011; Scheer et al., 2004). Because melatonin in our study was primarily administered for preventive rather than therapeutic purposes, we decided that a low-dose regimen would be the most appropriate. Although in most published clinical trials, the maximum olanzapine dose was 20 mg/day, both recent evidence and clinical experience (including our personal experience) justifies the use of up to 25 mg/day in the patients whose symptoms were inadequately controlled by 20 mg/day (Citrome and Kantrowitz, 2009). As an important example, in The Clinical Antipsychotic Trials of Intervention Effectiveness (CATIE) doses of up to 30 mg/day were allowed, with the mean dose of 20.1 mg/day, similar to what was administered in our study (Lieberman et al., 2005).

Our results must be interpreted in light of some limitations. Our study had small sample size which may explain "near-significant" or even "non-significant" *P* values in some of the secondary outcomes. We did not correct for multiple comparisons. The primary outcome was predefined and therefore would be unaffected by multiple comparisons. Regarding the secondary outcomes, because the study was the first in human, it can be best considered as exploratory rather than confirmatory. Therefore multiple comparisons can be left unadjusted (Bender and Lange, 2001). Also, we did not measure melatonin concentrations which could give us a quantitative measure of its effect on metabolic parameters. In addition, we did not control for diet and exercise after the patients were discharged from the hospital. However, randomization process minimizes the possibility of selection bias as an explanation for between-group differences. Therefore, even if diet and exercise variables were different, it could be due to the effect of melatonin on appetite and physical activity. Nevertheless, there is evidence that melatonin can cause weight loss independent of energy intake (Tan et al., 2011). Finally, although melatonin 3 mg/day was efficacious in short-term in patients with first episode schizophrenia, we cannot generalize its effect to other conditions, doses and longer treatment durations.

In summary, melatonin may be beneficial in short-term prevention of olanzapine-induced weight gain and hypertriglyceridemia in patients with first-episode schizophrenia. It can also improve psychotic symptoms when used as an adjunct to olanzapine.

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Contributors

A Modabbernia developed the concept. A Modabbernia, MJ Modabbernia, and AR Sobhani supervised the study process. PH Rad, MJ Modabbernia, and R Soleimani designed the study and wrote the protocol. M Ashrafi managed the literature search. PH Rad visited and followed the patients. ZA Roshan and S Taslimi undertook the statistical analysis. A Modabbernia, M Ashrafi, AR Sobhani wrote the first draft of the manuscript. All authors contributed to and have approved the final manuscript.

Conflict of interests

All authors report no financial relationships with commercial interests.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.jpsychires.2014.02.013.

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